

Name: _____

Due: September 23, 11:59PM via e-mail or Edwards Mailbox

1. What levels ℓj would result from the $n = 7$ harmonic oscillator state (where the first is $n = 0$)?
2. Other than nuclear masses, nuclear mass differences, radii, electric quadrupole moments, cosmic abundances, and deformation, name at least two other pieces of evidence for magic numbers in nuclei. Provide citation(s) for your answers.
3. Fill-out the single particle levels and, based on this, determine J^π for the ground-state and first two excited states of ^{29}Si . Explain how you came to your conclusions. Include both a drawing of the levels and the shorthand notation. Verify your results with the NNDC database.
4. Fill-out the single particle levels and, based on this, determine J^π for the ground-state and first two excited states of ^{34}Cl . Explain how you came to your conclusions. Include both a drawing of the levels and the shorthand notation. Verify your results with the NNDC database.
5. Within a single-particle model like the Shell Model, the Schmidt limit should provide an estimate for the nuclear magnetic moment of nuclei one particle outside of a closed shell. Plot the Schmidt limits and a sampling of the available experimental data for the ground state magnetic moments of odd proton nuclei (available at: <https://www-nds.iaea.org/nuclearmoments/>). You don't need to plot every nucleus but pick data for odd-Z systems over a range of spin and mass values. Comment on the resulting plot. Do you have any possible explanation for the deviation from the Schmidt limits?
6. The $E(2_1^+)$ and $E(4_1^+)$ for ^{226}Ra are 68 keV and 211 keV, respectively. Show that it is or is not a rigid rotor. Assuming it is, what is the expected value for $E(6_1^+)$? Compare to experiment.
7. ^{32}Mg has a deformation of $\beta = 0.51$. What are the expected excitation energies for the 2^+ , 4^+ , and 6^+ excited states, assuming it is a spheroidal rotor? Compare to the values from H. L. Crawford *et al.* Phys. Rev. C **93**, 031303(R) (2016).

8. Compare the ground-state J^π values for ^{173}Lu from the non-deformed and deformed shell-models to the experimental value from the NNDC.

